
CLICK HERE
CLICK HERE
APUL
NEET FLIX
AIM FREE EDUCATION TO ALL
NEET CHANNEL
JOIN OUR TELEGRAM CHANNELS HERE
TEST PAPERS CLICK
TOPPER'S NOTES CLICK
BEST FROM KOTA LECTURES CLICK
QUIZES CLICK
REVISION FLASH CARDS &
MANY MORE CLICK

PHYSICS

1)

On a planet particle is thrown with a velocity $\tilde{V} = 6\hat{i} + (20 - 4t)\hat{j}$. Then match the list.

List-I		List-II	
(P)	Time of flight	(1)	3
(Q)	Time when particle is moving at an angle of 53° with horizontal in upwards direction.	(2)	10
(R)	Range of the particle	(3)	50
(S)	Maximum height of the particle	(4)	60

(1) P → 2; Q → 1; R → 4; S → 3

(2) P → 1; Q → 2; R → 3; S → 4

(3) P → 2; Q → 3; R → 4; S → 1

(4) P → 2; Q → 4; R → 1; S → 3

2) **Assertion(A) :** The average speed of an object may be equal to arithmetic mean of individual speeds**.**
Reason(R) : Average speed is the ratio of total distance travelled and total time taken.

(1) Both assertion and reason are true and reason is the correct explanation of assertion

(2) Both assertion and reason are true but reason is not the correct explanation of assertion

(3) Assertion is true but reason is false

(4) Assertion is false but reason is true

3) If a person moves 4 km towards east then move 3 km towards north, after then he move 10 km towards 37° North of East. Magnitude of displacement of person in :

(1) 15 km

(2) 17 km

(3) 5 km

(4) 20 km

4) The velocity of a particle is given by $v = \sqrt{169 - 8x}$, where x is position in metre and v in m/s. The acceleration of the particle will be

(1) 7 m/s^2

(2) –4 m/s^2

(3) –8 m/s^2

(4) –2 m/s^2

5) A particle moves for 8 seconds. It first accelerates from rest and then retards to rest. If the retardation be 3 times the acceleration, then time for which it accelerates, will be :

(1) 2s

(2) 3s

(3) 4s

(4) 6s

6) The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at (4 + x) cm inside the block. The value of x is:

(1) 2.0

(2) 1.0

(3) 0. 5

(4) 1.5

7) A stone is dropped from a certain height which can reach the ground in 5 sec. It is stopped after three seconds of its fall and then is again released. The total time taken by the stone to reach the ground will be :-

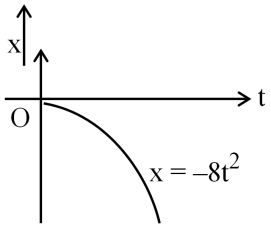
(1) 6 s

(2) 6.5 s

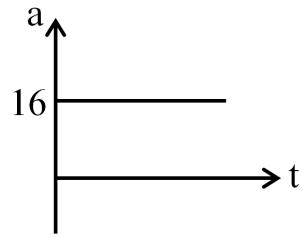
(3) 7 s

(4) 7.5 s

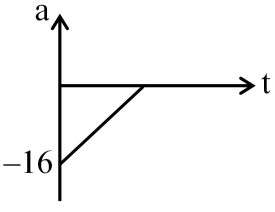
8) The displacement-time graph of a particle moving along a straight line (as shown) then acc-time graph is



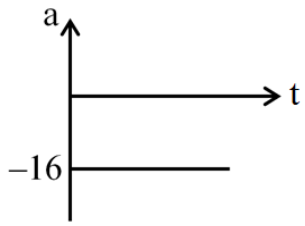
(1)



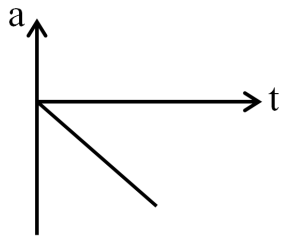
(2)



(3)

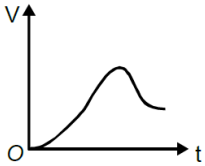


(4)

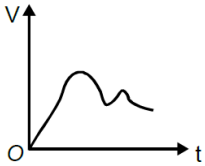


9) The following figures show some velocity V versus time t curves. But only some of these can be realised in practice. These are

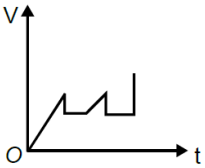
(a)



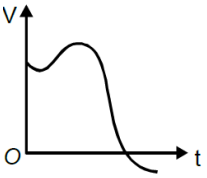
(b)



(c)



(d)



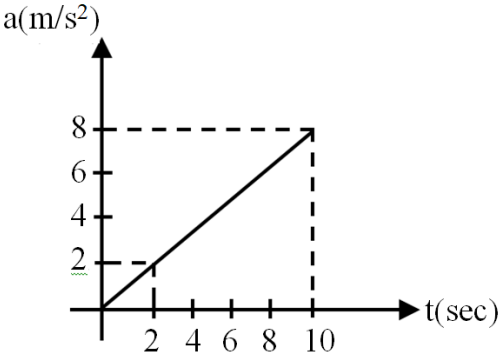
(1) Only a,b and d

(2) only a,b,c

(3) only b and c

(4) all of them

10) From the acceleration v/s time graph given below find the initial velocity of a body if its final velocity is 55 m/s -



(1) 15 m/s

(2) 16 m/s

(3) 17 m/s

(4) 18 m/s

11) A particle is projected with a velocity 120 m/s at an angle of 60° to the horizontal. The time (in

seconds) after which the particle will be moving upwards at an angle of 45° to the horizontal, will be ($g = 10\ m/s^2$)

(1) $6\left(\sqrt{3}-1\right)$

(2) $6-\sqrt{3}$

(3) $3\left(\sqrt{2}-1\right)$

(4) $6\left(\sqrt{3}+1\right)$

12) Two bodies are projected from ground with equal speeds 20 m/s from the same position in same vertical plane to have equal range but at different angle above the horizontal. If one of the angle is 30°, the sum of their maximum heights is: (assume $g = 10 m/s^2$)

(1) 400m

(2) 20m

(3) 30m

(4) 40m

13) The equation of a projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$ (where x & y is in metre and $g = 10\ m/s^2$). Find the maximum height attained by the projectile-

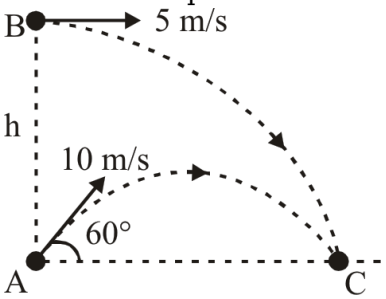
(1) $\frac{3}{10}$ m

(2) $\frac{3}{20}$ m

(3) $\frac{3}{40}$ m

(4) $\frac{3}{5}$ m

14) A particle A is projected from the ground with an initial velocity of 10 m/s at an angle of 60° with horizontal. From what height h should an another particle B be projected horizontally with 5 m/s, So that both the particles collide on ground at point C if both are projected simultaneously ?



(1) 10 m

(2) 30 m

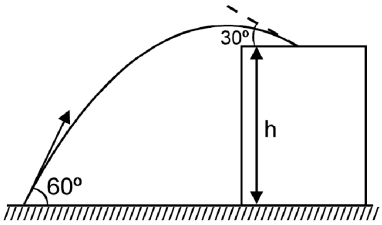
(3) 15 m

(4) 25 m

15) A ball is thrown from the roof of a building of height 44 m with speed V_0 at an angle θ below the horizontal. It lands 2 seconds later at a point 30 m from the base of the building. If $\tan\theta = \frac{x}{10}$ then, find the value of x ($g = 10 m/s^2$)

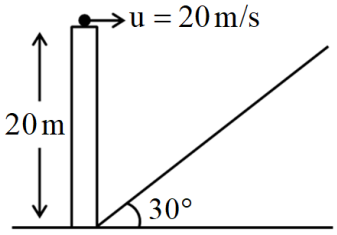
(1) 8
(2) 6
(3) 4
(4) 5

16) A stone projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height 'h'. Then the speed of projection of the stone is:



(1) $\sqrt{2gh}$
(2) $\sqrt{6gh}$
(3) $\sqrt{3gh}$
(4) $\sqrt{gh}$

17) A ball is projected horizontally such that it hits the plane shown in figure perpendicularly. Find the speed by which ball strikes with the plane.



(1) 40 m/s
(2) 20 m/s
(3) 30 m/s
(4) 15 m/s

18) A thief is running away on a straight road in a jeep moving with a speed of 9 m s^{-1}. A police man chases him on a motor cycle moving at a speed of 10 m s^{-1}. If the instantaneous separation of the jeep from the motorcycle is 100m, how long will it take for the police man to catch the thief ?

(1) 1s
(2) 19s
(3) 90s
(4) 100s

19) A man in a balloon, throws a stone downwards with a speed of 5 m/s with respect to balloon. The balloon is moving upwards with a constant acceleration of 5 m/s^2. Then velocity of the stone relative



to the man after 2 second is :

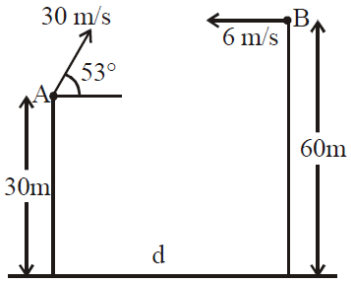
(1) 10 m/s

(2) 30 m/s

(3) 15 m/s

(4) 35 m/s

20) Ball A is projected from tower of height 30 m and B is projected horizontally at 6 m/s. A and B collide in air. If separation between the towers is d = 10x, then the value of x is :-



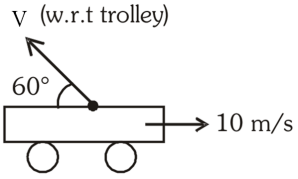
(1) 2m

(2) 3m

(3) 4m

(4) 6m

21) A particle is projected from a trolley with velocity v (w.r.t trolley) as shown in figure if an observer on ground observes that ball rises vertically. Then find maximum height reached by the ball.



(1) 10 m

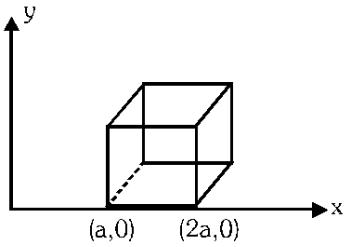
(2) 15 m

(3) 20 m

(4) 30 m

22) As shown in fig. the electric field in a region is given by $E = \frac{E_0}{a^2} x^2 \hat{(i)}$ volt/m then the total electric

flux through the cube of side 'a' will be :-



(1) Zero

(2) $3\ E_0\ a^2$

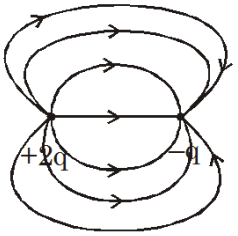
(3) E_0a^2

(4) $3E_0$

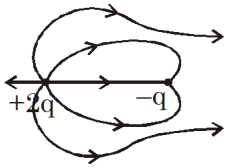
23)

The electric field line pattern for charges +2q & –q are shown. Select the most appropriate diagram :-

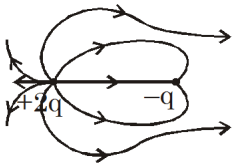
(1)



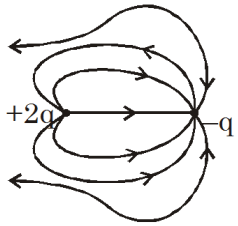
(2)



(3)



(4)



24) Two point charge of 100 μC and –4 μC are positioned at points $(-2\sqrt{3},\ 3\sqrt{3},\ -4)$ and $(4\sqrt{3},\ -5\sqrt{3},\ 6)$ respectively of a Cartesian coordinate system. Find the force vector on the –4 μC charge? All the coordinates are in meters.

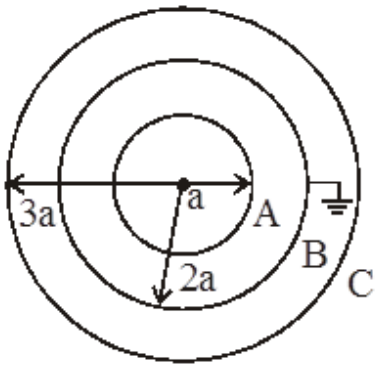
(1) $9 \times 10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$ N

(2) $9 \times 10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$ N

(3) $2.25 \times 10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$ N

(4) $2.25 \times 10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$ N

25) Figure shows a system of three concentric metal shells A, B and C with radii a, 2a and 3a respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell A, then the final charge on the shell B, is equal to :-



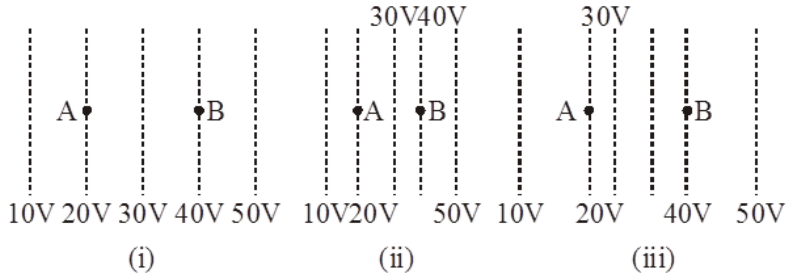
(1) $-\frac{4Q}{13}$

(2) $-\frac{8Q}{11}$

(3) $-\frac{5Q}{3}$

(4) $-\frac{3Q}{7}$

26) Figure shows some equipotential lines distributed in space. A charged object is moved from point A to point B:-



(1) The work done in figure (i) is the greatest.

(2) The work done in figure (ii) is the least.

(3) The work done is the same in figure (i), (ii) and (iii).

(4) The work done in figure (iii) is greater than figure (ii) but equal to that in figure (i).

27) A conducting sphere of radius R is given a charge Q. Consider three points A, B, C such that B at the surface, A at centre and C at a distance R/2 from the centre. The electric potential at these points are such that

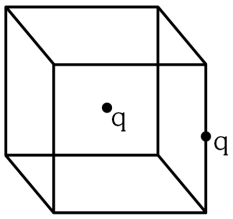
(1) $V_A = V_B = V_C$

(2) $V_A = V_B \neq V_C$

(3) $V_A \neq V_B \neq V_C$

(4) $V_A \neq V_B = V_C$

28) Two charges 'q' each are placed as shown in figure. Then flux passing through the cube is –



(1) $\frac{3}{2}\left(\frac{q}{\varepsilon_0}\right)$

(2) $\frac{5}{4}\left(\frac{q}{\epsilon_0}\right)$

(3) $\frac{2}{3}\left(\frac{q}{\epsilon_0}\right)$

(4) $\frac{5}{12}\left(\frac{q}{\epsilon_0}\right)$

29) Two point charge A and B have charge 0.5 C and 2C respectively and their position vectors are ($\hat{i}+\hat{j}+\hat{k}$)m and ($-\hat{i}-\hat{j}+3\hat{k}$) m respectively. The force on the charge at A due to B is –

(1) $\frac{k(\hat{i}+\hat{j}-\hat{k})}{12\sqrt{3}}$N

(2) $\frac{k}{24\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$ N

(3) $\frac{k}{24\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ N

(4) $\frac{k}{24\sqrt{3}}(-\hat{i}-\hat{j}+\hat{k})$ N

30) A point charge q is placed in a cubical box of side a. If all the dimensions of box are changed by a factor of 3 then by what factor the flux crossing the box will change :-

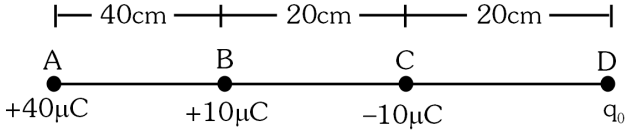
(1) Decrease by 9 times

(2) Increased by 9 times

(3) Either (1) or (2)

(4) Doesn't Change

31) In given diagram, find value of q_0 so that +10μC remain in equilibrium :



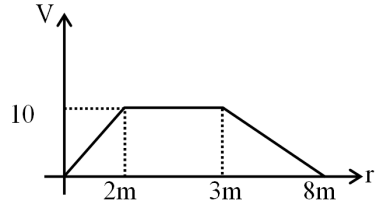
(1) –80 μC

(2) +40 μC

(3) +80 μC

(4) –20 μC

32) In the following graph, the magnitude of maximum electric field is :



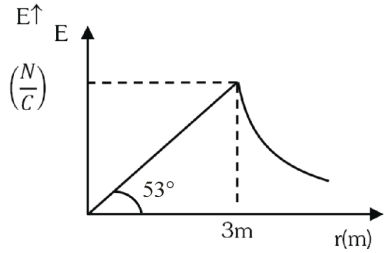
(1) $5Vm^{-1}$

(2) $2.5\ Vm^{-1}$

(3) $1.25\ Vm^{-1}$

(4) $10\ Vm^{-1}$

33) For a solid non – conducting sphere, electric field variation with distance from the centre is shown. Then net charge in the sphere is (In coulomb) –



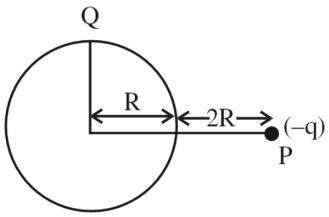
(1) $180\ \pi\in_0$

(2) $162\ \pi\in_0$

(3) $144\ \pi\in_0$

(4) $72\pi\in_0$

34) Electric potential energy of charge (–q) is –



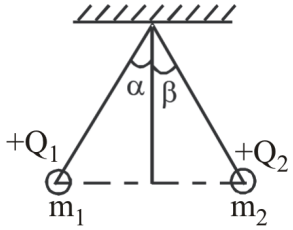
(1) $\frac{kQq}{2R}$

(2) $\frac{kQq}{R}$

(3) $\frac{-kQq}{3R}$

(4) $\frac{kQq}{3R}$

35) Consider the following charged suspended ball system. If $\alpha < \beta$, then the statements which can



be true at equillibrium out of followings are :- (a) $Q_1 > Q_2$, $m_1 < m_2$ (b) $Q_1 > Q_2$, $m_1 > m_2$
(c) $Q_1 < Q_2$, $m_1 = m_2$ (d) $Q_1 < Q_2$, $m_1 > m_2$

(1) Only d

(2) Only c

(3) (c, d)

(4) (b, d)

36) The electric potential at a point (x, y, z) is given by $V = -x^2y - xz^3 + 4$. The electric field $\vec{E}$ at that point is:

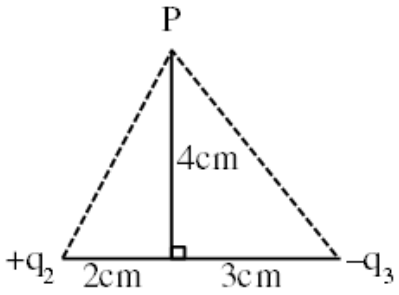
(1) $\vec{E} = \hat{i}2xy + \hat{j}(x^2 + y^2) + \hat{k}(3xz - y^2)$

(2) $\vec{E} = \hat{i}z^3 + \hat{j}xyz + \hat{k}z^2$

(3) $\vec{E} = \hat{i}(2xy - z^3) + \hat{j}xy^2 + \hat{k}3z^2x$

(4) $\vec{E} = \hat{i}(2xy + z^3) + \hat{j}x^2 + \hat{k}3xz^2$

37) If the net electric field at point P along Y axis is zero, then the ratio of $\frac{q_2}{q_3}$ is $\frac{8}{5\sqrt{x}}$, where x = ...



(1) 5

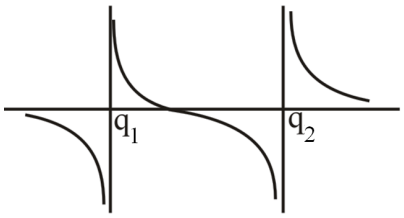
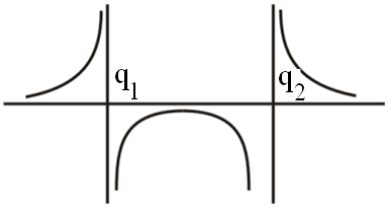
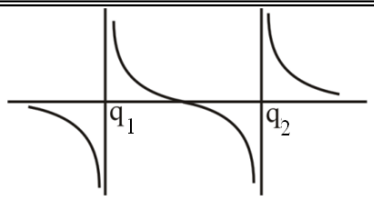
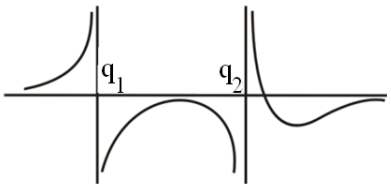
(2) 6

(3) 7

(4) 8

38) Variation of electric field along the line joining two charges is given in list-I. Relation between charges is given in list-II. Match the proper variation. Take electric field along right as positive

List-I	List-II

(P)		(1)	$\|q_1\| > \|q_2\|$ $q_1 < 0; q_2 > 0$
(Q)		(2)	$\|q_1\| = \|q_2\|$ $q_1 > 0; q_2 > 0$
(R)		(3)	$\|q_1\| = \|q_2\|$ $q_1 < 0; q_2 > 0$
(S)		(4)	$\|q_2\| > \|q_1\|$ $q_1 > 0; q_2 > 0$

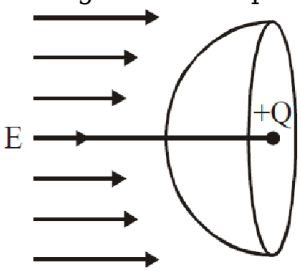
(1) (P) - (3) ; (Q) - (1) ; (R) - (2) ; (S) - (4)

(2) (P) - (1) ; (Q) - (4) ; (R) - (2) ; (S) - (3)

(3) (P) - (4) ; (Q) - (3) ; (R) - (2) ; (S) - (1)

(4) (P) - (2) ; (Q) - (3) ; (R) - (4) ; (S) - (1)

39) The figure shows a point charge +Q placed at the center of an imaginary hemispherical surface of radius R. A uniform electric field E is applied in the horizontal direction. If the net flux passing through the hemispherical surface is zero then what is the magnitude of the electric field E.



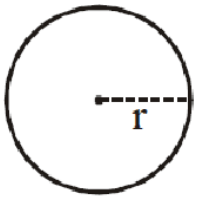
(1) $\frac{Q}{2\pi\varepsilon_0 R^2}$

(2) $\frac{Q}{2\varepsilon_0 R^2}$

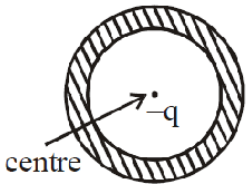
(3) $\frac{Q}{\pi\varepsilon_0 R^2}$

(4) $\frac{Q}{\varepsilon_0 R^2}$

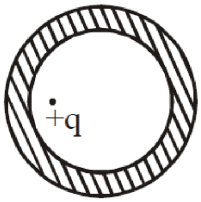
40) In which of the case(s) we will get uniform charge distribution on external spherical surface. Given every object is a conductor.

(A) 

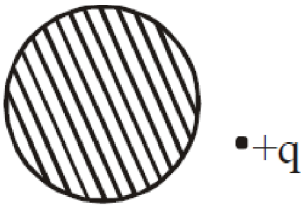
thin shell having net charge +q

(B) 

thick shell having net charge zero

(C) 

thick shell having net charge zero

(D) 

solid sphere having net charge +q

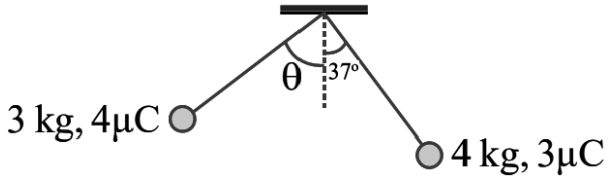
(1) Only (A) & (B)

(2) Only (A), (B) & (C)

(3) Only (A)

(4) All

41) Two small positively charged spheres are suspended from a common point at the ceiling by the insulating light strings of equal length as shown. If one string makes an angle 37° with the vertical, find the angle θ at equilibrium.



(1) 30°

(2) 37°

(3) 53°

(4) 60°

42) A small sphere carrying a charge 'q' is hanging in between two parallel plates by a string of length L. Time period of pendulum is T_0. When parallel plates are charged, the time period changes to T. The ratio T/T_0 is equal to :

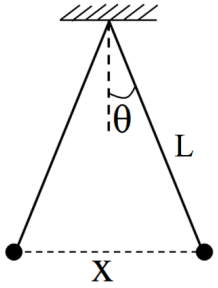
(1) $\left(\frac{g+\frac{qE}{m}}{g}\right)$

(2) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{3/2}$

(3) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{1/2}$

(4) None of these

43) In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L. Assume that θ is so small that tanθ = sinθ, then for equilibrium x is equal to :



(1) $\left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{1/3}$

(2) $\left(\frac{qL^2}{2\pi\varepsilon_0 mg}\right)^{1/3}$

(3) $\left(\frac{q^2L^2}{4\pi\varepsilon_0 mg}\right)^{1/3}$

(4) $\left(\frac{q^2L}{4\pi\varepsilon_0 mg}\right)^{1/3}$

44) **Case-1 :** Five identical point charges of magnitude Q are kept at the corners of a regular pentagon of centre to corner distance a.
Case-2 : One charge is now removed. Match the following for above two cases :

	Column-I		**Column-II**
(A)	Electric field at the centre of pentagon in case-1	(P)	$\frac{1}{4\pi\varepsilon_0}\frac{Q\times 5}{a}$
(B)	Electric potential at the centre of pentagon in case-1	(Q)	Zero
(C)	Electric field at the centre of pentagon in case-2	(R)	$\frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2}$
(D)	Electric potential at the centre of pentagon in case-2	(S)	$\frac{1}{4\pi\varepsilon_0}\frac{Q}{a}\times 4$

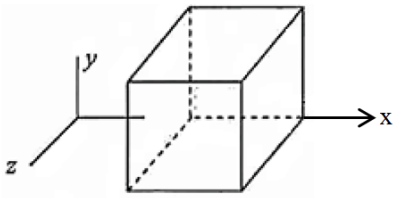
(1) (A → Q); (B → P); (C → R); (D → S)

(2) (A → P); (B → Q); (C → S); (D → R)

(3) (A → S); (B → Q); (C → P); (D → R)

(4) (A → R); (B → Q); (C → P); (D → S)

45) The figure shows a cube of side with edges parallel to the axes. The electric field is in positive y-direction but its magnitude changes such that it is $\frac{15}{\varepsilon_0 \ell^2}$ at the bottom face and $\frac{5}{\varepsilon_0 \ell^2}$ at the top face in SI unit. Find the charge enclosed by the cube.



(1) 5 C
(2) 10 C
(3) –10 C
(4) Zero

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**2**	**1**	**2**	**4**	**3**	**3**	**3**	**1**	**1**	**1**	**2**	**2**	**3**	**1**	**3**	**1**	**4**	**4**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**2**	**2**	**2**	**2**	**2**	**3**	**1**	**2**	**1**	**4**	**3**	**1**	**3**	**3**	**4**	**4**	**1**	**3**	**1**	**2**
Q.	41	42	43	44	45															
A.	**3**	**3**	**1**	**1**	**3**															

SOLUTIONS

PHYSICS

1) $\tilde{v} = 6\hat{i} + (20 - 4t)\hat{j}$

time when $v_y = 0$

$20 - 4t = 0$

$t = 5$

$T = 2 \times t = 10$ sec

$\tan 53° = \dfrac{20 - 4t}{6}$

$\dfrac{4}{3} = \dfrac{20 - 4t}{6}$

$t = 3$ sec

$R = v_x \times T$

$= 6 \times 10 = 60$ m

$$H = \frac{v_y^2}{2 \times a} = \frac{20 \times 20}{2 \times 4} = \frac{400}{8} = \frac{200}{4} = 50\text{m}$$

2) average speed $V_{av.} = \dfrac{v_1 t_1 + v_2 t_2 + v_3 t_3 + ...}{t_1 + t_2 + t_3 + ...}$

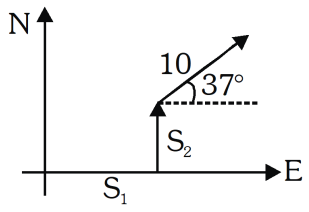
If $t_1 = t_2 = t_3 = ... = t$

Then $V_{av.} = \dfrac{(v_1 + v_2 + v_3 + ...)\,t}{nt}$

or $V_{av.} = \dfrac{v_1 + v_2 + v_3 + ...}{n}$

It means that, average speed may be equal to arithmetic mean of individual speeds.

3)



$\vec{S}_{net} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3 = 4\hat{i} + 3\hat{j} + (8\hat{i} + 6\hat{j})$

$\vec{S}_{net} = 12\hat{i} + 9\hat{j}$

$\left|\vec{S}_{net}\right| = 15$

4) $v = \sqrt{169 - 8x}$

$v^2 = 169 - 8x$

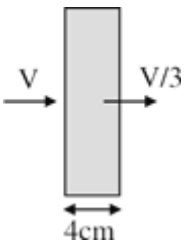
Comparing with $v^2 = u^2 + 2as$

$u = 13$ m/s and $a = -4$ m/s^2

5) $v = \alpha t_1 = \beta t_2 \Rightarrow \frac{\alpha}{\beta} = \frac{t_2}{t_1}$

If $\beta = 3\alpha$, then $t_1 = 3t_2$

Since $t_1 + t_2 = 8$sec, hence $t_1 = 6$sec.



6)

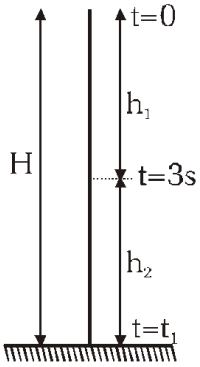
$$\left(\frac{V}{3}\right)^2 = V^2 - 2a(4) \Rightarrow a = \frac{8V^2}{9(8)} = \frac{V^2}{9}$$

$$0 = V^2 - 2a(4 + x)$$

$$\Rightarrow \quad V^2 = 2\left(\frac{V^2}{9}\right)(4 + x)$$

$4.5 = 4 + x$

$x = 0.5$



7) $H = \frac{1}{2}g(5)^2 = \frac{1}{2}(10)(5)^2 = 125\text{m}$

$h_1 = \frac{1}{2}g(3)^2 = 45\text{m}$

$h_2 = H - h_1 = 80\text{m}$

But $h_2 = \frac{1}{2}g(t_1 - 3)^2$

So, $80 = \frac{1}{2}(10)(t_1 - 3)^3$

$\Rightarrow t_1 - 3 = 4 \Rightarrow t_1 = 7\text{s}$

8) $V = \frac{dx}{dt} = -16t$ & $a = \frac{dv}{dt} = -16\ \text{m/s}^2$

9)

In graph (c) there is sudden change in velocity, which is practically not possible so only a, b & d can be realised in practice.

10) ΔV = Area under a - t graph

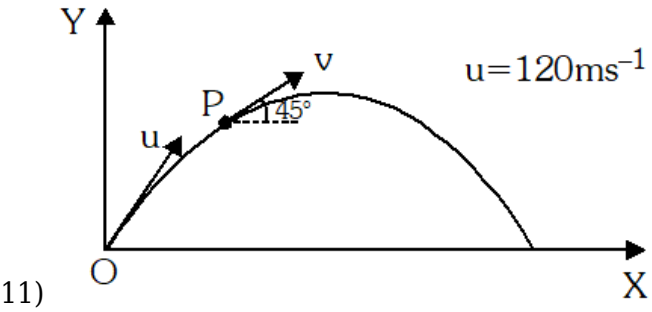
$V - u = \frac{1}{2}(2 + 8) \times 8$

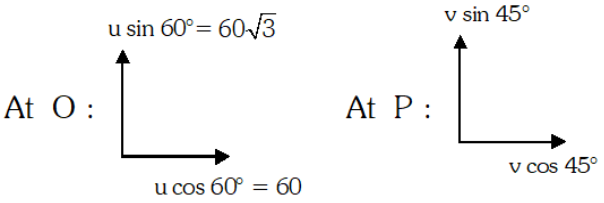
$55 - u = 40$

$u = 55 - 40$

$= 15$ m/s

11)





Horizontal component of velocity = constant

$\Rightarrow 60 = v \cos 45°$

$v = 60\sqrt{2}$

Now $v_y = u_y + a_y t$

$\Rightarrow v \sin 45° = 60\sqrt{3} - 10t$

$\Rightarrow 60 = 60\sqrt{3} - 10t$

$\boxed{t = 6(\sqrt{3} - 1)}$

12)

For same range

$\theta_1 = \theta$ then $\theta_2 = 90° - \theta$

$\theta_1 = 30° \Rightarrow \theta_2 = 60°$

$H_1 + H_2 = \frac{u^2 \sin^2 30°}{2g} + \frac{u^2 \sin^2 60°}{2g}$

$= \frac{u^2}{2g}\left(\frac{1}{4} + \frac{3}{4}\right) = \frac{u^2}{2g}$

$= \frac{20 \times 20}{2 \times 10} = 20$ m

13)

Explain Question : Equation of tragectory is given of a projectile & we need to calculate its maximum height.

Concept : This question is based on projectile motion.

Solution :

$y = x\tan\theta\left(1-\frac{x}{R}\right) = \sqrt{3}x\left[1-\frac{g\,x}{2\sqrt{3}}\right]$

$\theta = 60°$, $R = \frac{2\sqrt{3}}{g}$

$\frac{R}{H} = \frac{4}{\tan\theta}$

$H = \frac{R\tan\theta}{4} = \frac{2\sqrt{3}}{g} \times \frac{\sqrt{3}}{4} = \frac{3}{20}$ m

Final Answer : option (2)

14) R must same

$\sqrt{\frac{2h}{g}} \times 5 = \frac{2\,(10\cos 60^o)\,(10\sin 60^o)}{g}$

h = 15 m

15) $x = v_0\cos\theta\, t$

$y = 30 - v_0\cos\theta \times 2$

$\Rightarrow v_0\cos\theta = 15$

$\Rightarrow S_y = u_y t + \frac{1}{2}a_y t^2$

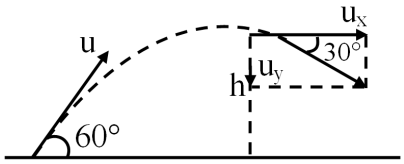
$\Rightarrow 44 = v^0\sin\theta \times 2 + \frac{1}{2} \times 10 \times (2)^2$

$\Rightarrow v_0\sin\theta = 12$

$\Rightarrow \tan\theta = \frac{12}{15} = \frac{x}{10}$ $\Rightarrow x = 8$

16)

Explain Question : Asking about the speed of projection for the given condition in diagram.
Concept : This question is based on collision of projectile with wall.
Explanation :



$\tan 30^\circ = \frac{|u_y|}{|u_x|}$

$\tan 30^\circ = \frac{\sqrt{u_y^2 - 2g(y)}}{|u_x|}$

$\because u_x = u_x$

$\tan 30° = \frac{\sqrt{u^2\sin^2 60^\circ - 2gh}}{u\cos 60^\circ}$

on solving $u = \sqrt{3gh}$

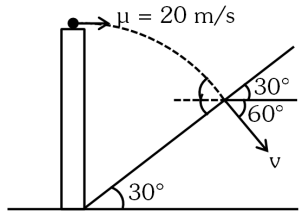
Final answer : 4

17)

Explain Question : Find speed by which ball strikes with plane as shown in figure.

Concept : This question is based on Projection from height [Horizontal Projection]

Solution :

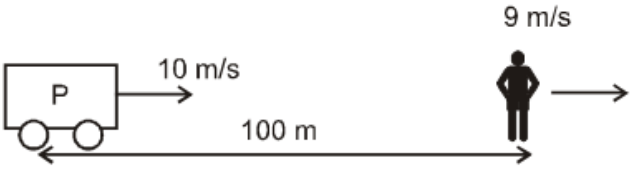


$v \cos 60° = 20$

$\boxed{v = 40 \text{ m/s}}$

Final Answer : option (1)

18)



$\vec{V}_{PT} = \vec{V}_P - \vec{V}_T = 10 - 9 = 1 \text{ m/s}$

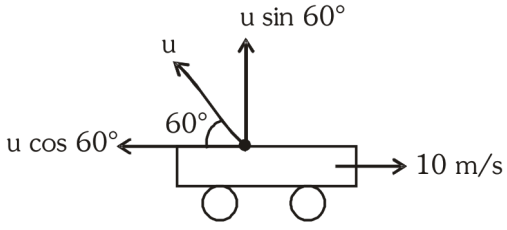
So time taken $t = \frac{100}{\vec{V}_{PT}} = \frac{100}{1} = 100$ sec.

19)

Relative velocity of stone = 5 m/s
relative acceleration of stone = 10 + 5 = 15 m/s^2
v = u + at = 5 + 15 × 2 = 35 m/s
relative velocity after t = 2 second is 35 m/s

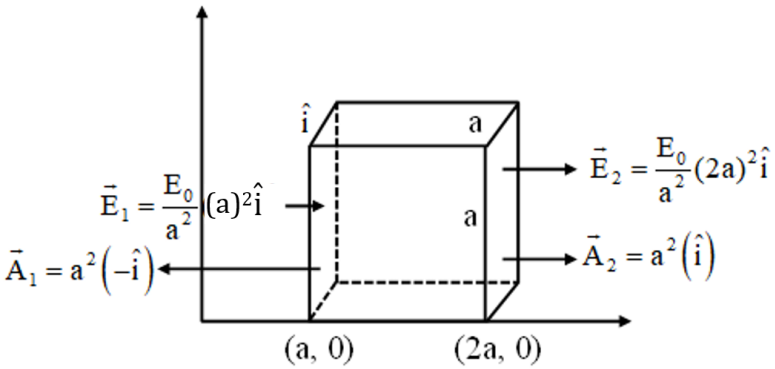
20) Relative motion along Y-axis implies
60–30 = (30 sin 53°)t ⇒ t = 1.25 s
Relative motion along x-axis implies
d = (30 cos 53° + 6) 1.25 = 30 m

21)



As balls makes vertically upwards then
u cos 60° = 10 ⇒ 20 m/s
So maximum height reached by ball is :

$H_{max} = \frac{(u \sin 60^\circ)^2}{2g} = \frac{(10\sqrt{3})^2}{20} = \frac{100 \times 3}{20} = 15$ m



22)

$\phi = \vec{E}_1 \cdot \vec{A}_1 + \vec{E}_2 \cdot \vec{A}_2$

$\phi = E_0(\hat{i}).a^2(-\hat{i}) + \frac{E_0}{a^2}(4a^2)\hat{i}.a^2(\hat{i})$

$\phi = -E_0 a^2 + 4E_0 a^2$

$\phi = 3E_0 a^2$

23)

Field lines are proportional to quantity of charge & they originate at positive charge & terminates at –ve charge or at ∞

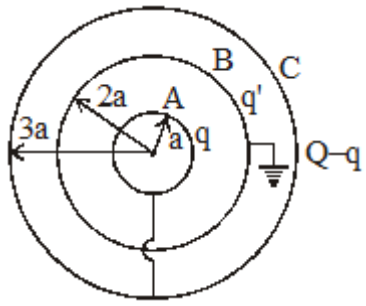
24) (1) (2)

100 μc –4μc

$(2\sqrt{3}, 3\sqrt{3}, -4)$ $(4\sqrt{3}, -5\sqrt{3}, 6)$

$\vec{F}_{21} = \frac{kq_1q_2\vec{r}_{12}}{|\vec{r}_{12}|^3}$

25) $V_B = 0$



$\frac{kq}{2a} + \frac{kq'}{2a} + k\frac{(Q-q)}{3a} = 0$ ...(i)

$V_A = V_C$

$\frac{kq}{a} + \frac{kq'}{2a} + \frac{k(Q-q)}{3a} = \frac{k(q+q'+Q-q)}{3a}$ ...(ii)

from (i) & (ii)

$q' = \frac{-8Q}{11}$

26) In all the three figures, V_A = 20 V and V_B = 40 V
Work done in carrying a charge q from A to B is
$W = q\,(V_B - V_A)$
Hence, work done is same in all figure.

27)

In electrostatic condition potential inside a conductor is constant and equal to the potential at the surface.

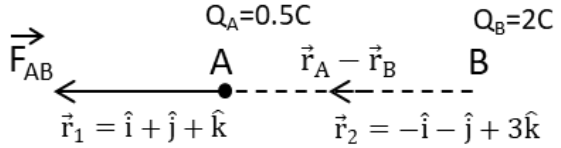
28)

Flux of cube centred charge is $\frac{q}{\epsilon_0}$ and flux of edge centred charge is $\frac{q}{4\epsilon_0}$.
Total flux through the cube
$= \frac{q}{\epsilon_0} + \frac{q}{4\epsilon_0} = \frac{5}{4}\left(\frac{q}{\epsilon_0}\right)$

29)

The **vector form of the Coulomb force** on charge **A** due to another point charge **B** is:

$$\vec{F}_{AB} = \frac{kQ_AQ_B}{|r_{AB}|^3}\left[\vec{r}_A - \vec{r}_B\right]$$



$$\vec{F}_{AB} = \frac{K\,(0.5)\,(2)\left[\left(\hat{i}+\hat{j}+\hat{k}\right) - \left(-\hat{i}-\hat{j}+3\hat{k}\right)\right]}{\left|\sqrt{(1-(-1))^2 + (1-(-1))^2 + (-2)^2}\right|^3}$$

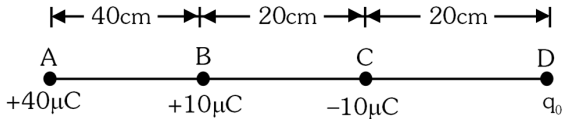
$$\vec{F}_{AB} = \frac{K\,(1)}{\left[\sqrt{12}\right]^3}\left[2\hat{i} + 2\hat{j} - 2\hat{k}\right]$$

$$\vec{F}_{AB} = \frac{2K}{12 \times 2\sqrt{3}}\left[\hat{i} + \hat{j} - 2\hat{k}\right]$$

$$\vec{F}_{AB} = \frac{K\left(\hat{i}+\hat{j}-\hat{k}\right)}{12\sqrt{3}}$$

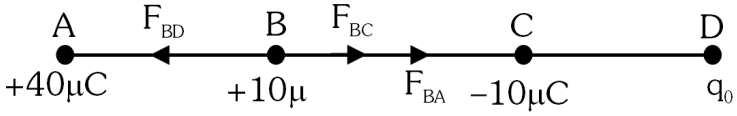
30) Flux does not depend on size.

31)



For equilibrium,

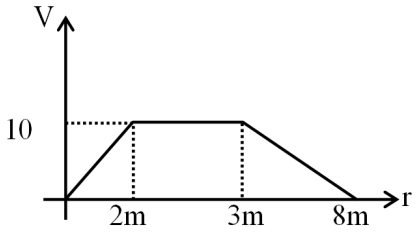
$F_{BD} = F_{BC} + F_{BA}$



$$\frac{K10 \times q_0}{(40)^2} = \frac{K10 \times 10}{(20)^2} + \frac{K10 \times 40}{(40)^2}$$

$$\frac{q_0}{(40)^2} = \frac{10}{(20)^2} + \frac{40}{(40)^2}$$

$$q_0 = 10 \times \left(\frac{40}{20}\right)^2 + 40 = +80\ \mu C$$



32)

$$E = -\frac{dv}{dr} \Rightarrow |E| = \left|\frac{dv}{dr}\right| \Rightarrow |E| = (\tan\theta)$$ [slope of V-r curve]

From r = 0 to r = 2m

$$|E_1| = |\tan\theta| = \left|\frac{10}{2}\right| = 5 \text{ V/m}$$

from r = 2m to r = 3m

θ= 0º (line parallel to x-axis)

$$|E_2| = (\tan\theta) = \tan 0^o = 0$$

for r = 3 m to r = 8m

$$|E_3| = |\tan\theta| = \left|\frac{10}{5}\right| = 2 \text{ V/m}$$

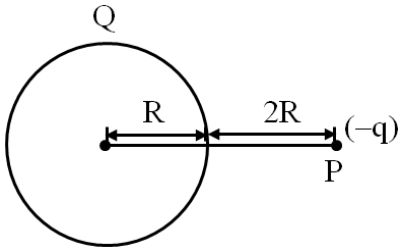
So the maximum magnitude of E is 5 V/m

33) Inside a uniformly charged sphere, the linear region has slope $\tan 53° = \frac{4}{3}$. With radius R = 3m, the surface field is $E(R) = \frac{4}{3} \cdot 3 = 4$ N/C. Using

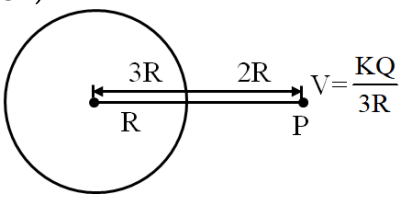
$$E(R) = \frac{1}{4\pi\varepsilon_0 R^2} \Rightarrow Q = 4\pi\varepsilon_0 E(R) R^2$$

we get, substituting E = 4 and R = 3

$Q = 4\pi\varepsilon_0 \cdot 4 \cdot 9 = 144\ \pi\varepsilon_0$ coulomb b



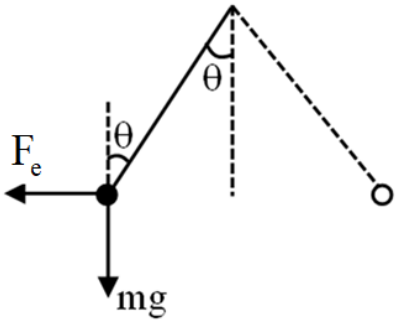
34)



Potential energy at point P = –q × V

$$= -q \times \frac{KQ}{3R}$$

$$= -\frac{KQq}{3R}$$



35)

$$\tan\theta = \frac{F_e}{mg}$$

$$\tan\theta = \frac{KQ_1Q_2}{r^2mg}$$

$$\theta \propto \frac{1}{m}$$

as $\alpha < \beta \quad m_1 > m_2$

So, $Q_1 > Q_2$ & $m_1 > m_2$

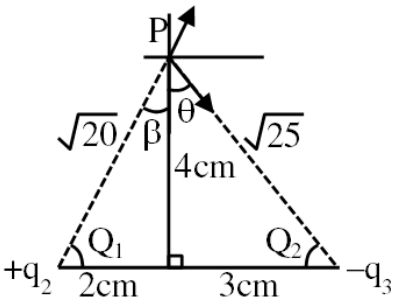
$Q_1 < Q_2$ & $m_1 > m_2$

36)

$$\vec{E} = -\left(\frac{\partial v}{\partial x}\hat{i} + \frac{\partial v}{\partial y}\hat{j} + \frac{\partial v}{\partial z}\hat{k}\right)$$

$$= -\left[\left(-2xy - z^3\right)\hat{i} - x^2\hat{j} - 3xz^2\hat{k}\right]$$

$$= \left(2xy + z^3\right)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$$



37)

$$\frac{Kq_2}{20}\cos\beta = \frac{Kq_3}{25}\cos\theta$$

$$\frac{Kq_2}{20}\frac{4}{\sqrt{20}} = \frac{Kq_3}{25}\frac{4}{\sqrt{25}}$$

$$\frac{q_2}{q_3} = \frac{20}{25}\sqrt{\frac{20}{25}} = \frac{8}{5\sqrt{x}}$$

$$\Rightarrow \sqrt{x} = \frac{8\times 25\sqrt{25}}{5\times 20\sqrt{20}}$$

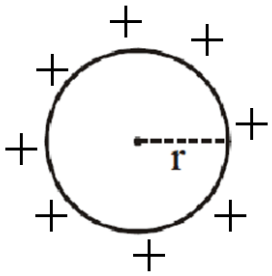
x = 5

38) Apply the following tests :-
(1) Electric field will be zero in between the charges only if the charges are of same nature.
(2) E.F. will be zero in between the charges, closer to the smaller charge
(3) E.F. may be zero on outer side of charges (on the line joining them), if they are unequal, closer to the smaller charge.

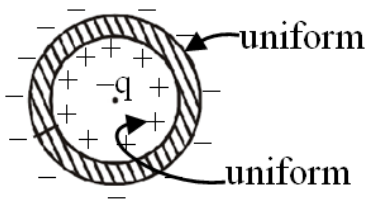
39) $E \times (\pi R^2) = \frac{Q}{2\varepsilon_0}$

$$E = \frac{Q}{2\pi R^2 \varepsilon_0}$$

40)

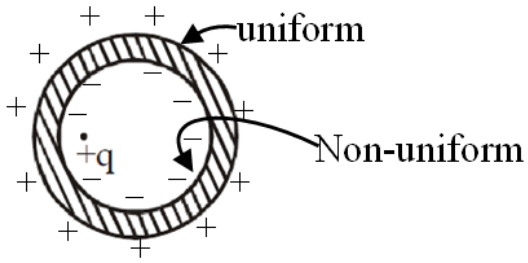


(A)
+q charge is uniformly distributed at outer surface

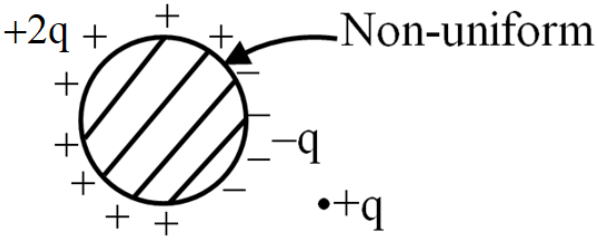


(B)

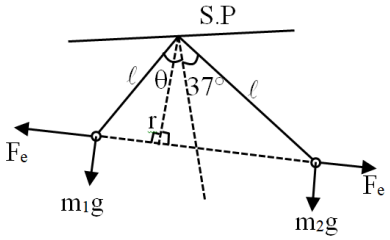
(C)



(D)



41)



Here system is in equilibrium masses are not equal so line joining masses is not horizontal or angle made by strings with vertical are different. Here net torque on m_1 and m_2 is zero. Here we draw a perpendicular from suspension point to line joining charges. Length of this perpendicular consider as r. Let magnitude of electrostatic force between charge particles is F_e. Now balance the torque on m_1 about suspension point (S.P.)

$F_e\, r = m_1 g \sin\theta$ ...(1)

for m_2 $F_e\, r = m_2 g \sin 37°$ ...(2)

by eq^n (1) & (2)

$$\sin\theta = \frac{m_2}{m_1}\sin 37^\circ = \frac{4}{3}\times\frac{3}{5} = \frac{4}{5}$$

Q = 53°

42) $$T_0 = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}} = 2\pi\sqrt{\frac{\ell}{\left(g+\frac{qE}{m}\right)}}$$

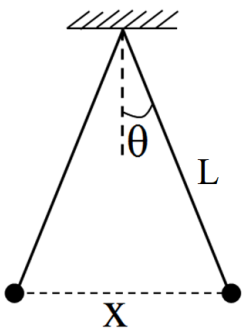
$$\frac{T}{T_0} = \left(\frac{g}{g+\frac{qE}{m}}\right)^{\frac{1}{2}}$$

43) $\tan\theta = \frac{Fe}{mg}$ $\because \theta << \Rightarrow \tan\theta \approx \sin\theta$

$x^3 = \frac{2kq^2L}{mg}$

$x = \left(\frac{2kq^2L}{mg}\right)^{1/3}$

$\sin\theta = \frac{kq^2}{x^2mg} = \frac{x}{2L}$



44) Point charges are uniformly distributed around the centre O. Hence electric field E is zero.

 (A) → Q.

Electric potential $V = 5 \times \left(\frac{Q}{4\pi\varepsilon_0 a}\right)$

 (B) → P

In case 2, the electric field of three identically placed charges is zero. Net field at the centre of O is due to a single point charge and is given as

$E = \frac{Q}{4\pi\varepsilon_0 a^2}$

 (C) → R

In case 2, the electric potential, being scalar, becomes

$V = 4 \times \frac{Q}{4\pi\varepsilon_0 a}$

 (D) → S

45) Flux enclosed by the loop,

$\phi = \frac{-15}{\varepsilon_0 \ell^2}(\ell^2) + \frac{5}{\varepsilon_0 \ell^2}(\ell^2) = \frac{-10}{\varepsilon_0} = \frac{Q_{in}}{\varepsilon_0}$

So, Q_{in} = –10 C